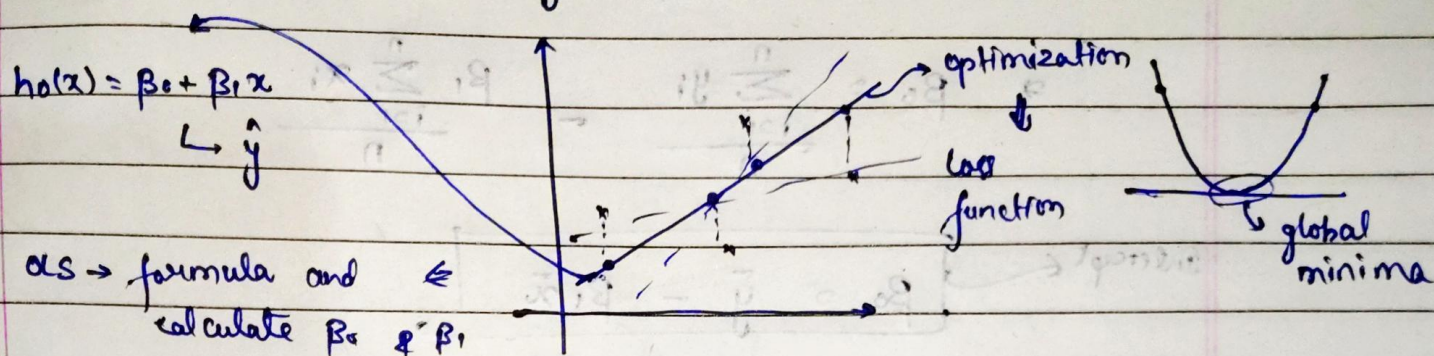


* Linear Regression using OLS: (Ordinary Least Square).



Ordinary Least Square: Error (MSE)

$$S(\beta_0, \beta_1) = \frac{1}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

$\Rightarrow$ Aim β_0 & β_1

$$\frac{\partial S}{\partial \beta_0}(\beta_0, \beta_1) = \frac{2}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) (0 - 1) = 0$$

$$\Rightarrow -\frac{2}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0 \quad \text{--- (I)}$$

$$\frac{\partial S}{\partial \beta_1}(\beta_0, \beta_1) = \frac{2}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) (-x_i) = 0$$

$$\Rightarrow -\frac{2}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) (x_i) = 0 \quad \text{--- (II)}$$

Eq. (I) $-\frac{2}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0$

$$\Rightarrow -\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0$$

$$\Rightarrow -\sum_{i=1}^n y_i + n\beta_0 + \beta_1 \sum_{i=1}^n x_i = 0$$

$$\Rightarrow n\beta_0 + \beta_1 \sum_{i=1}^n x_i = \sum_{i=1}^n y_i$$

$$\Rightarrow \beta_0 = \frac{\sum_{i=1}^n y_i}{n} - \beta_1 \frac{\sum_{i=1}^n x_i}{n}$$

Intercept $\leftarrow \beta_0 = \bar{y} - \beta_1 \bar{x}$

Eq. (1) $-\frac{2}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)(x_i) = 0$

$$\Rightarrow \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)(x_i) = 0$$

$$\Rightarrow \sum_{i=1}^n x_i y_i - \beta_0 \sum_{i=1}^n (x_i) - \beta_1 \sum_{i=1}^n (x_i)^2 = 0$$

① $\Rightarrow \sum_{i=1}^n (x_i y_i - \beta_0 x_i - \beta_1 x_i^2) = 0$
 $\hookrightarrow$ Replace $\beta_0 = \bar{y} - \beta_1 \bar{x}$

$$\Rightarrow \sum_{i=1}^n (x_i y_i - (\bar{y} - \beta_1 \bar{x}) x_i - \beta_1 x_i^2) = 0$$

$$\Rightarrow \sum_{i=1}^n (x_i y_i - x_i \bar{y} + \beta_1 \bar{x} x_i - \beta_1 x_i^2) = 0$$

$$\Rightarrow \sum_{i=1}^n (y_i - \bar{y} + \beta_1 \bar{x} - \beta_1 x_i) x_i = 0$$

$$\Rightarrow \sum_{i=1}^n [(y_i - \bar{y}) + \beta_1 (\bar{x} - x_i)] = 0$$

$$\Rightarrow \sum_{i=1}^n (y_i - \bar{y}) + \sum_{i=1}^n \beta_1 (\bar{x} - x_i) = 0$$

$$\Rightarrow \sum_{i=1}^n \beta_1 (\bar{x} - x_i) = - \sum_{i=1}^n (y_i - \bar{y})$$

$$\beta_1 = \frac{- \sum_{i=1}^n (y_i - \bar{y})}{\sum_{i=1}^n (\bar{x} - x_i)}$$

$$\Rightarrow \beta_1 = \frac{- \sum_{i=1}^n (y_i - \bar{y})}{- \sum_{i=1}^n (x_i - \bar{x})}$$

$$\Rightarrow \left[\beta_1 = \frac{\sum_{i=1}^n (y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})} \right] \Rightarrow \text{Coefficient}$$

coefficient Intercept
 $\searrow$ $\searrow$

x	y	$(y_i - \bar{y}) \div (x_i - \bar{x}) \rightarrow \beta_1$	$\beta_0 = (\bar{y} - \beta_1 \bar{x})$
'	'	'	'
'	'	'	'
'	'	'	'
'	'	'	'
$\bar{x}$	$\bar{y}$		

OLS $\approx$ Linear Regression (sk learn)